

$$\mathcal{E} = -\frac{d\Phi}{dt} = \oint \mathbf{E} \cdot d\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} = IR$$

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a}$$

Electromag
HW # 3
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Problem 1

Same flux (Φ) passes through every turn of both coils due to iron core "concentrating" the flux. *** If current (I) is changing show that emf ($\mathcal{E}$) in the secondary is: $\left[\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_2}{N_1}\right]$ ***
 $\mathcal{E}_1$ is the back emf of the primary.

Both coils experience the same flux. Lets still treat them separately for now.

Magnetic field inside a solenoid is $[B = \mu_0 n I]$. n is the # of turns per unit length $\rightarrow [n = \frac{N}{l}]$. Lets build the emf for both coils:

Primary $\rightarrow N_1$ turns $\rightarrow B_1 = \mu_0 \frac{N_1}{l} I(t) \rightarrow \Phi = \int \mathbf{B} \cdot d\mathbf{a} \rightarrow d\mathbf{a} = \pi r^2$
 $\rightarrow \Phi_1 = \int (\mu_0 \frac{N_1}{l} I(t)) \cdot (\pi r^2) dr \rightarrow \Phi_1 = \mu_0 \frac{N_1}{l} I \pi \int r^2 dr$
 $\rightarrow [\mu_0 \frac{N_1}{l} I \pi \frac{r^3}{3} = \Phi_1] \rightarrow \mathcal{E}_1 = -\frac{d\Phi}{dt} \rightarrow \left[\mathcal{E}_1 = \frac{\mu_0 N_1 I \pi r^3}{3l}\right]$

$\rightarrow$ Taking derivative of $\mathcal{E}_1$ w respect to time at one instance we have

$\rightarrow$ Okay I felt like I was on the right track and that I messed up somewhere. Turns out I did. $\rightarrow [\Phi_1 = \mu_0 n_1 \pi r^2 I(t)]$

$\rightarrow$ or $[\Phi_1 = \mu_0 n I(t) A] \rightarrow$ Referencing equation (7.22) on page 322 in 4th edition we have $\rightarrow \mathcal{E}_1 = -N_1 \frac{d\Phi}{dt}$ where N_1 is simply

the M term in equation (7.22). $\rightarrow -N_1 (\mu_0 \frac{N_1}{l} \pi r^2 I(t)) \frac{dI}{dt}$
 $\rightarrow \left[-\left(\frac{\mu_0 N_1^2 \pi r^2}{l}\right) \frac{dI}{dt} = \mathcal{E}_1\right]$

Secondary $\rightarrow$ Since the flux is the same through both loops

$\rightarrow \mathcal{E}_2 = -N_2 \frac{d\Phi}{dt} \rightarrow \mathcal{E}_2 = -N_2 (\mu_0 \frac{N_1}{l} \pi r^2 I(t)) \frac{dI}{dt}$

$\rightarrow \mathcal{E}_2 = -\left(\frac{\mu_0 N_1 N_2 \pi r^2}{l}\right) \frac{dI}{dt}$

Putting the equations in the solution form or setting them equal

$\rightarrow \frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{-\left(\frac{\mu_0 N_1 N_2 \pi r^2}{l}\right) \frac{dI}{dt}}{-\left(\frac{\mu_0 N_1^2 \pi r^2}{l}\right) \frac{dI}{dt}}$

The only thing left is $\mathcal{E} \propto N$
 $\frac{N_1}{N_2} = \frac{1}{N_1} -$

$\rightarrow \left[\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_2}{N_1}\right]$